

Practical No. 9

Aim: To implement join-like operations using \$lookup in MongoDB.

Output:

```
> db.students.aggregate([{$lookup: { from: "departments", localField: "dept", foreignField: "dept", as: "deptInfo"}}])
< {
  _id: 1,
  name: 'Sam Altman',
  dept: 'CSE',
  deptInfo: [
    {
      _id: ObjectId('69bd62c43f87e52649b32243'),
      dept: 'CSE',
      hod: 'Eren Yeager'
    }
  ]
}
{
  _id: 2,
  name: 'Jeff Bezos',
  dept: 'AI',
  deptInfo: [
    {
      _id: ObjectId('69bd62c43f87e52649b32244'),
      dept: 'AI',
      hod: 'Light Yagami'
    }
  ]
}
{
  _id: 3,
  name: 'Elon Musk',
  dept: 'CSE',
  deptInfo: [
    {
      _id: ObjectId('69bd62c43f87e52649b32243'),
      dept: 'CSE',
      hod: 'Eren Yeager'
    }
  ]
}
```

```
> db.students.aggregate([{$lookup: { from: "courses", localField: "course_id", foreignField: "course_id", as: "course_details"}}])
< {
  _id: 1,
  name: 'Sam Altman',
  dept: 'CSE',
  course_id: 201,
  course_details: [
    {
      _id: ObjectId('69bd62ee3f87e52649b32245'),
      course_id: 201,
      course_name: 'Computer Science'
    }
  ]
}
{
  _id: 2,
  name: 'Jeff Bezos',
  dept: 'IT',
  course_id: 202,
  course_details: [
    {
      _id: ObjectId('69bd62ee3f87e52649b32246'),
      course_id: 202,
      course_name: 'Information Technology'
    }
  ]
}
{
  _id: 3,
  name: 'Elon Musk',
  dept: 'CSE',
  course_id: 201,
  course_details: [
    {
      _id: ObjectId('69bd62ee3f87e52649b32245'),
      course_id: 201,
      course_name: 'Computer Science'
    }
  ]
}
}
```

Commands:

```
db.students.aggregate([{$lookup: { from: "departments", localField: "dept", foreignField: "dept", as:
"deptInfo"}}])
```

```
db.students.aggregate([{$lookup: { from: "courses", localField: "course_id", foreignField: "course_id",
as: "course_details"}}])
```